One-Dimensional Newtonian Dynamics

Review

- **Force**: a push or pull on an object from another object
- **Sum of forces on an object**: $\vec{F}_{\text{net}} = \sum_{i=1}^{N} \vec{F}_i = \vec{F}_1 + \vec{F}_2 + \ldots + \vec{F}_N$
- Newton's laws of motion:

1. A body remains at rest, or in motion at constant velocity (speed and direction), unless acted on by a net force
2. At any instant of time, the net force on a body is equal to the body's acceleration multiplied by its mass, $\vec{F}_{\text{net}} = m\vec{a}$
3. Forces (generally) come in pairs: the force of body A on body B and the force of B on A have the same magnitude but opposite direction, $\vec{F}_{A \text{ on } B} = -\vec{F}_{B \text{ on } A}$

Using Newton's second law

Problem solving strategy

- Model object as particle.
- Apply simplifications
- Draw a picture, establish a coordinate system, define symbols, identify what the problem wants
- Identify all forces acting on the object at that instant
- Make a free-body diagram
- Either use Newton's second law to find acceleration, then use kinematics to find velocity and position or use kinematics to find acceleration, then solve for unknown forces.

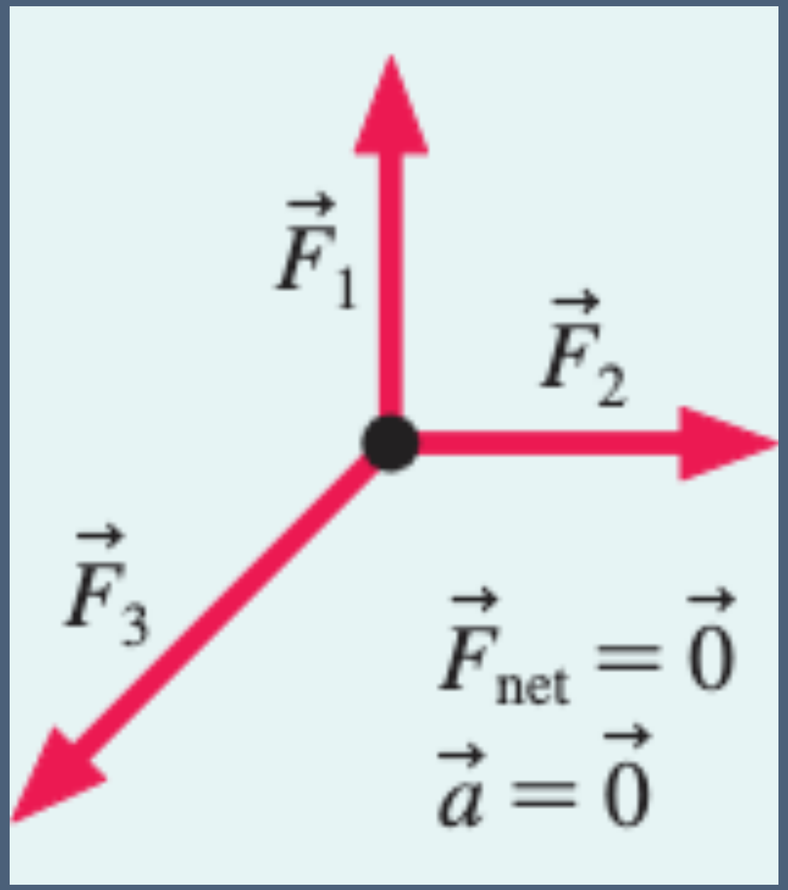
Identifying forces

Every force needs a source, so look for the sources rather than the forces:

- Pick the object of interest; the thing you're analyzing (only forces exerted on it count)
- Start with gravity (it's the only force in this class that acts without touching)
- Go around the object and list everything touching it; each thing touching exerts a force
  - A surface gives a normal force (and possibly a friction force)
  - A rope or string gives tension (pulling along the rope)
  - A spring gives a spring force (pushing if compressed, pulling is stretched)
  - A hand or other object gives a push or pull
  - Air or water gives drag (often ignored)

Mechanical equilibrium model

- When $\vec{F}_{\text{net}} = 0$ object at rest or moves with constant velocity ("equilibrium" or "forces balanced")
- From Newton's second law $\vec{F}_{\text{net}} = m\vec{a} = 0$, i.e., $\vec{a} = 0$
- If forces aren't balanced $\vec{F}_{\text{net}} \neq 0$ and model cannot be applies
  - i.e., particle moving with acceleration $\vec{a}$



$$\vec{F}_{\text{net}} = (F_{\text{net}})_x\hat{i} + (F_{\text{net}})_y\hat{j} = \sum_i (F_i)_x\hat{i} + \sum_i (F_i)_x\hat{j}$$

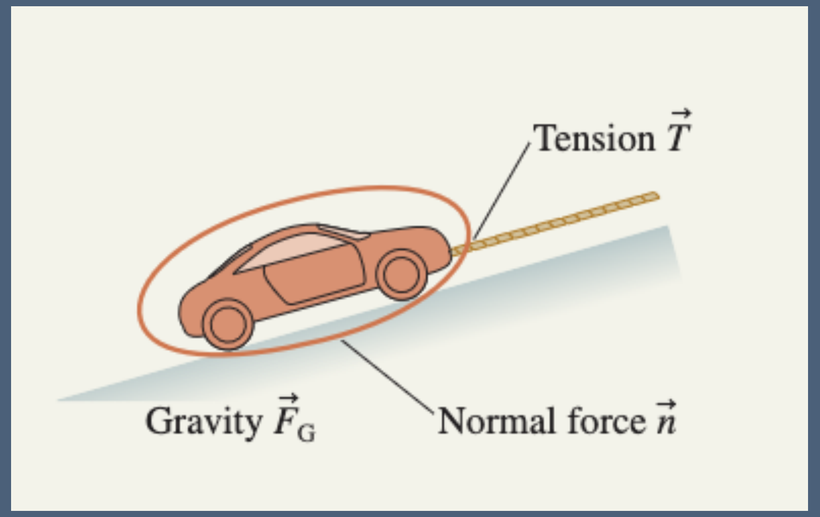
Example

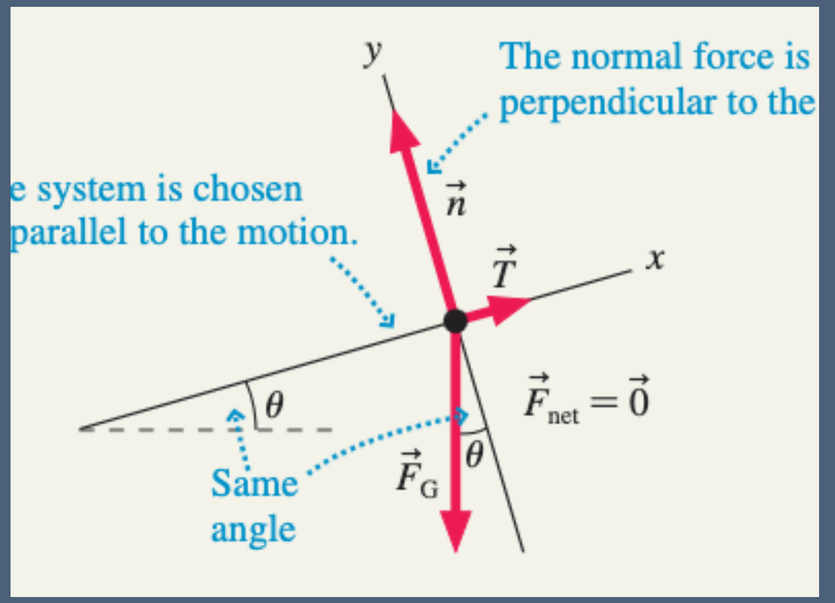
- A car with a weight of $15{,}000\ \mathrm{N}$ is being towed up a $20°$ slope at constant velocity. Friction is negligible. The tow rope is rated at $6000\ \mathrm{N}$ maximum tension. Will it break?

- Draw a picture, establish a coordinate system, define symbols, identify what the problem wants
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- Either use $\vec{F}_{\mathrm{net}}$ to find $\vec{a}$, then use kinematics or use kinematics to find $\vec{a}$, then solve for unknown forces.

Example

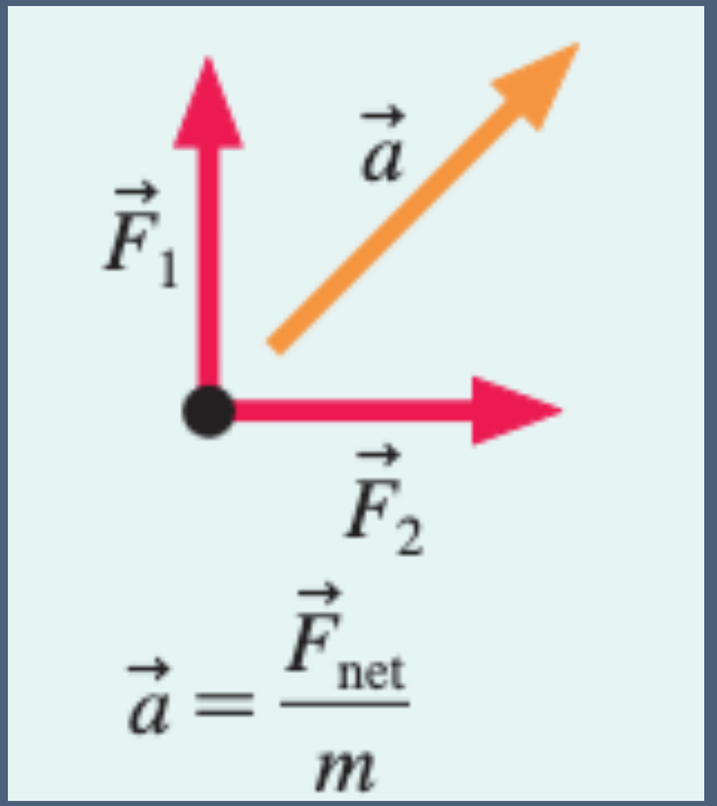
- A car with a weight of $15{,}000\ \mathrm{N}$ is being towed up a 20° slope at constant velocity. Friction is negligible. The tow rope is rated at $6000\ \mathrm{N}$ maximum tension. Will it break?
- $T = 5100\ \mathrm{N}$, since $T < 6000\ \mathrm{N}$ the rope will not break





Constant force model

- If $\vec{F}_{\text{net}} =$ constant, then all forces are constant and $\vec{a} =$ constant ("uniform acceleration")
- From Newton's second law $\vec{F}_{\text{net}} = m\vec{a} =$ constant
- If $\vec{a}$ uniform, we can use our kinematic equations.
- Model fails if any of the forces aren't constant.



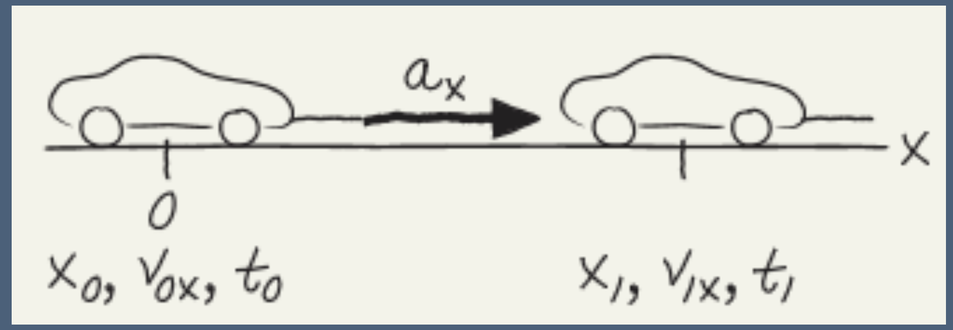
Example

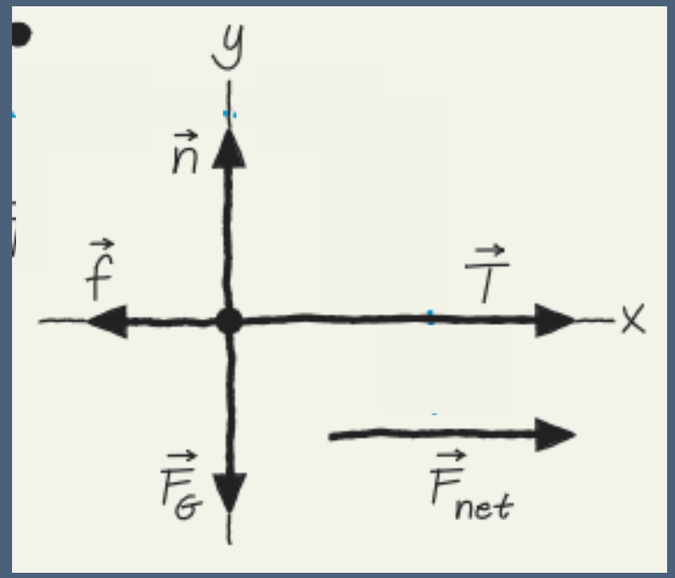
- A $1500\,\mathrm{kg}$ car is pulled by a tow truck. The tension in the tow rope is $2500\,\mathrm{N}$, and a $200\,\mathrm{N}$ friction force opposes the motion. If the car starts from rest, what is the speed after $5.0\,\mathrm{s}$?

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Example

- A $1500\,\mathrm{kg}$ car is pulled by a tow truck. The tension in the tow rope is $2500\,\mathrm{N}$, and a $200\,\mathrm{N}$ friction force opposes the motion. If the car starts from rest, what is the speed after $5.0\,\mathrm{s}$?
- $a_x = 1.53\,\mathrm{m/s^2}$
- $v_1 = 7.7\,\mathrm{m/s}$



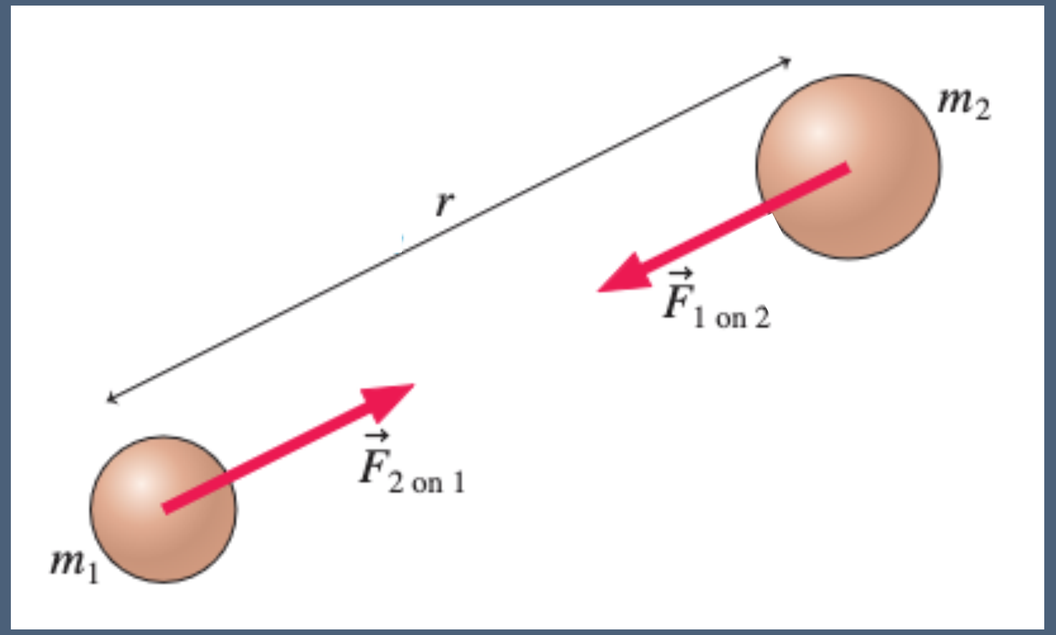


Mass

- Scalar quantity that describes the amount of matter in an object AND an object's inertia
- Mass is an intrinsic property (i.e., depends on the object itself)
- SI units: $\mathbf{kg}$

Gravity force

- An attractive, long-range force between two objects
- Force given by Newton's law of gravity
  $|\vec{F}_{1\text{ on }2}| = |\vec{F}_{2\text{ on }1}| = G\frac{m_1 m_2}{r^2}$ where
  $G = 6.67 \times 10^{-11}\,\mathrm{N\,m^2/kg^2}$ is the gravitational constant
- Gravity isn't constant, it gets weaker with distance
- Human size objects have tiny gravity (for 2 humans of mass $70\,\mathrm{kg}$ separated by $1\,\mathrm{m}$ about $4.23\,\mu\mathrm{N}$, the weight of your text book is about $13.3\,\mathrm{N}$).
- At the scale of planets or larger gravity is significant



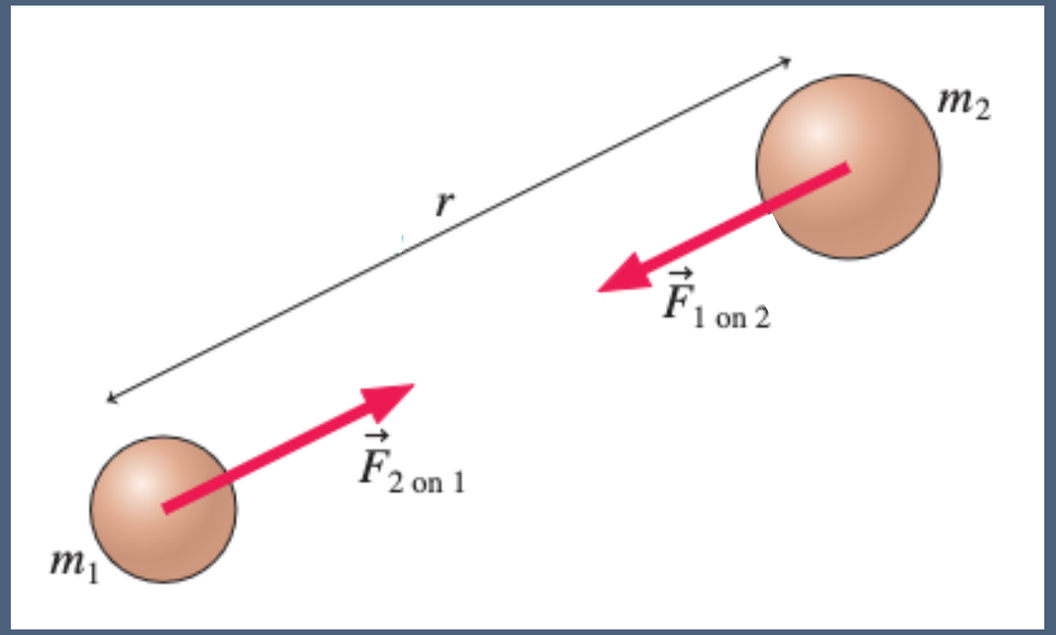
Gravity force

- For objects near the surface of a planet we assume there's virtually no difference between r and the planet's radius R
- The gravitational force of the planet on mass m is approximated as

$$|\vec{F}_G| = G\frac{Mm}{R^2}$$

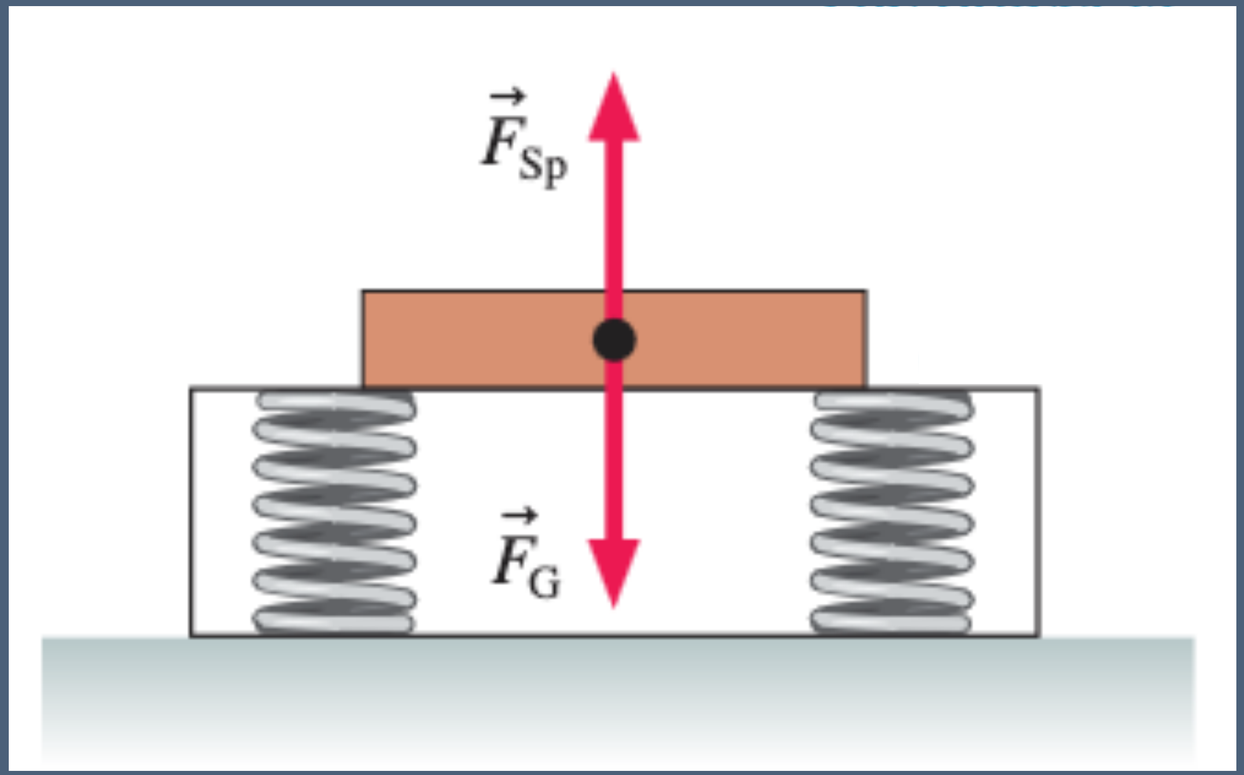
- We define $g = G\frac{M}{R^2}$. so

$$|\vec{F}_G| = F_G = mg$$



Weight

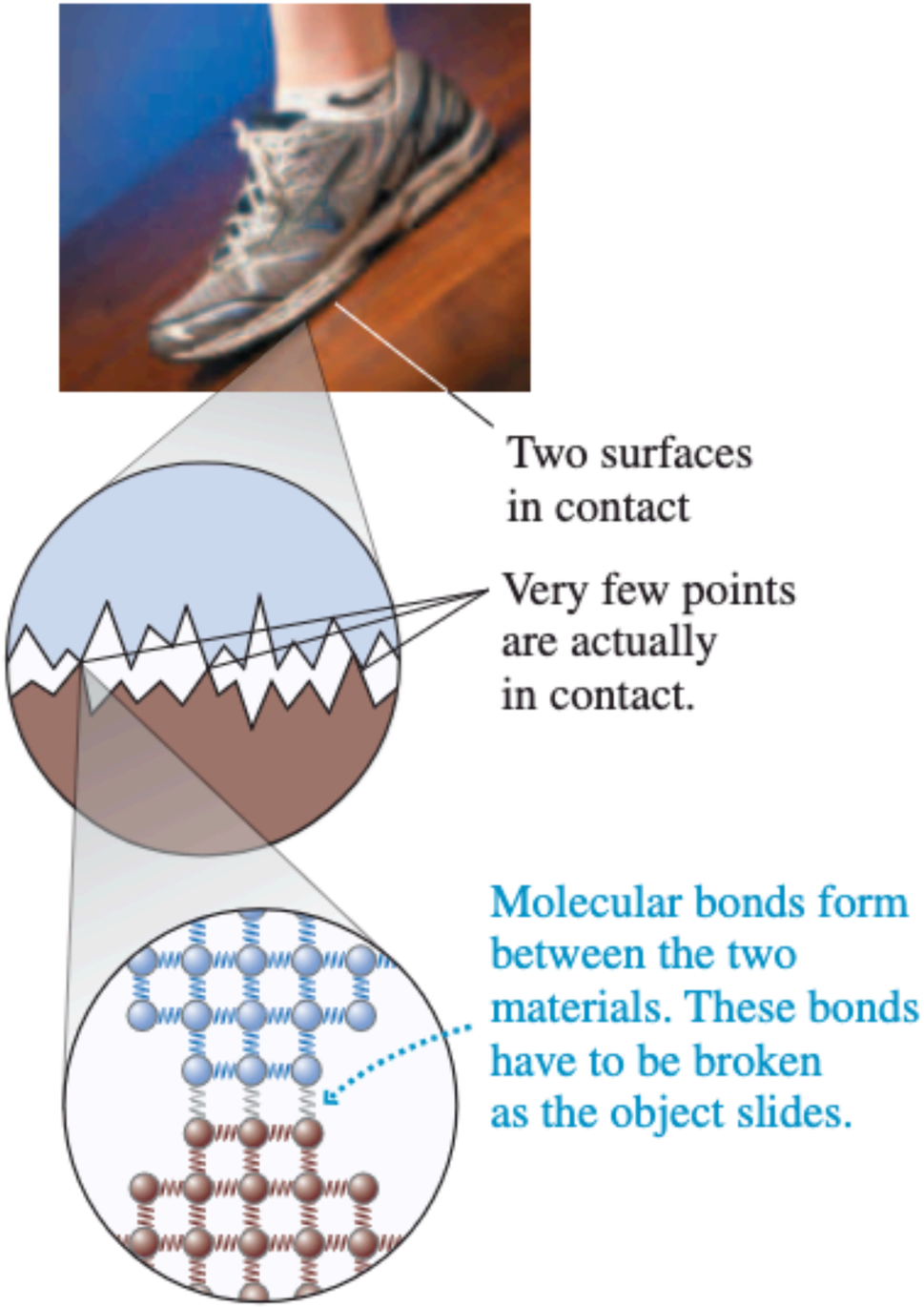
- Extrinsic property (depends on the object's surroundings or interaction with other objects)
- SI unit: $\mathbf{N}$
- Physics of weight
  - Weigh yourself by standing on a spring scale
  - When on the scale you're stationary, so $\vec{F}_{\text{net}} = 0$, which means $\vec{F}_{Sp} = -\vec{F}_{G}$ and since $F_G = mg$, $\vec{F}_{Sp} = -mg$.
- Weight is a force with magnitude $w = mg$



Friction

Physical picture

- All surfaces are rough at microscopic scales, so they touch only at their peaks.
- Pushing the surfaces together harder increases the contact area. This is why friction depends on the normal force, n.
- Molecular bonds form at the contact points. These cause static friction. To make an object slip, you must push hard enough to break the bonds.
- Once the surfaces are sliding, the bonds don't have time to fully form. This is why kinetic friction is smaller than static friction.
- A lubricant (oil, air) reduces friction by keeping the surfaces apart, so there are fewer contact points.

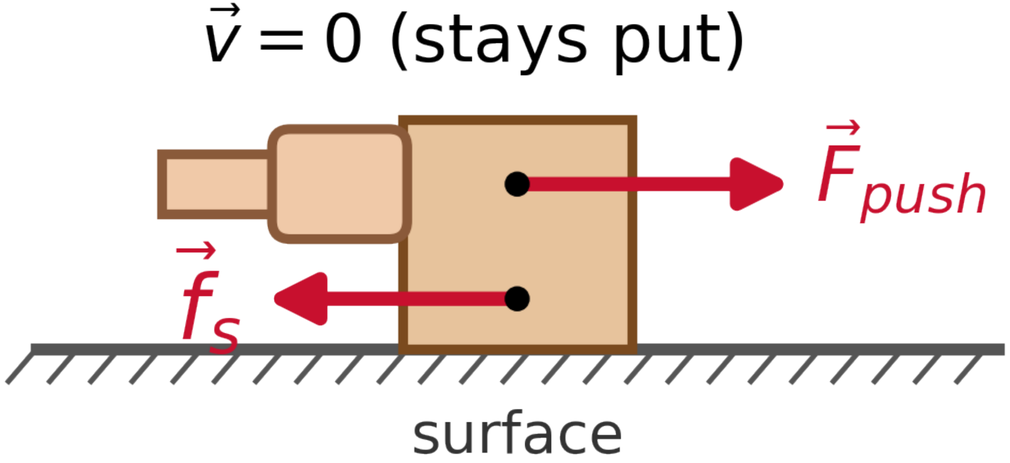


Physics tutoring

- PS 104 (lounge)
- M, W 1:00PM – 2:30PM; T, Th 2:30PM – 4:00PM
- Tutor: David

Static friction

- f_s : force that keeps an object from slipping
- Object not moving so forces in equilibrium
- Parallel with surface, points in the direction which prevents motion
- Size of static friction depends on how hard the object is pushed or pulled
- $f_{s\,\max} = \mu_s n$
  - maximum force required to cause object to slip (depends on surface); true static friction equals amount needed to maintain equilibrium
  - μ_s coefficient of static friction (dimensionless)



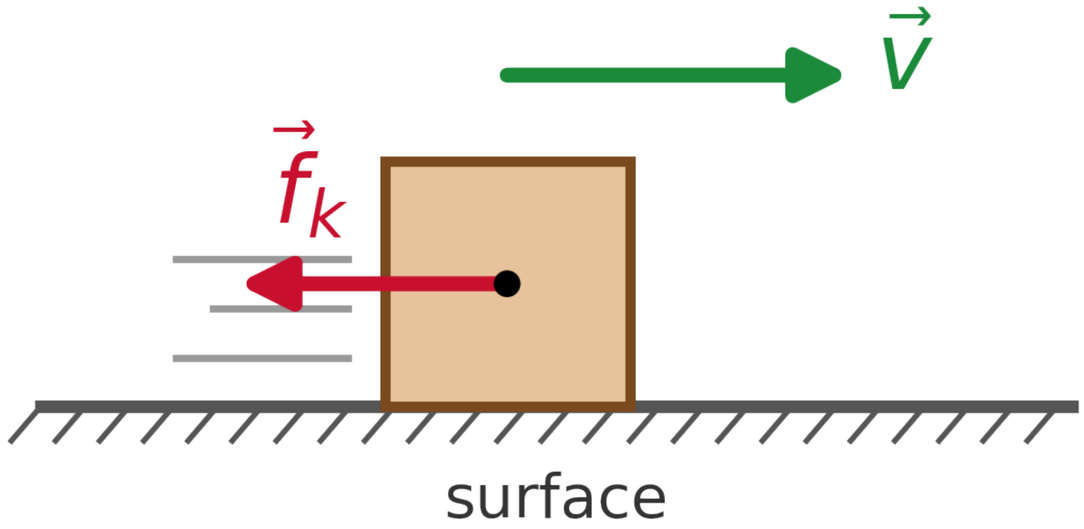
$f_s < f_{s\,\max}$ object remains at rest

$f_s = f_{s\,\max}$ object slips

$f_s > f_{s\,\max}$ not physically possible

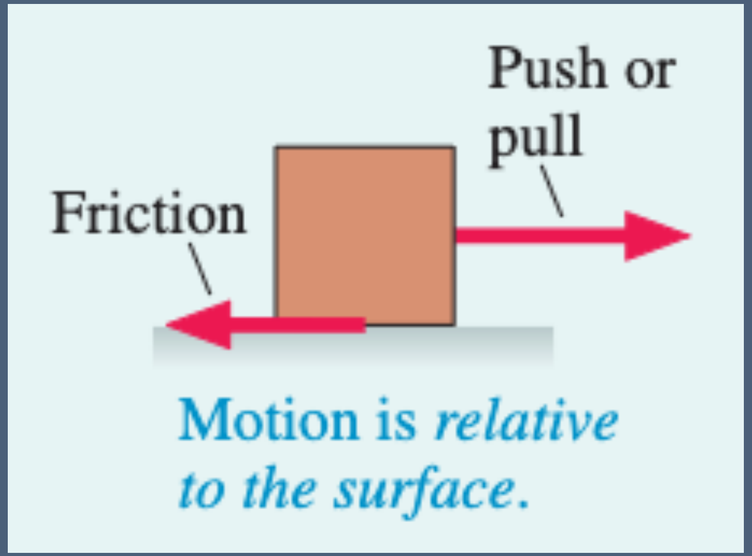
Kinetic friction

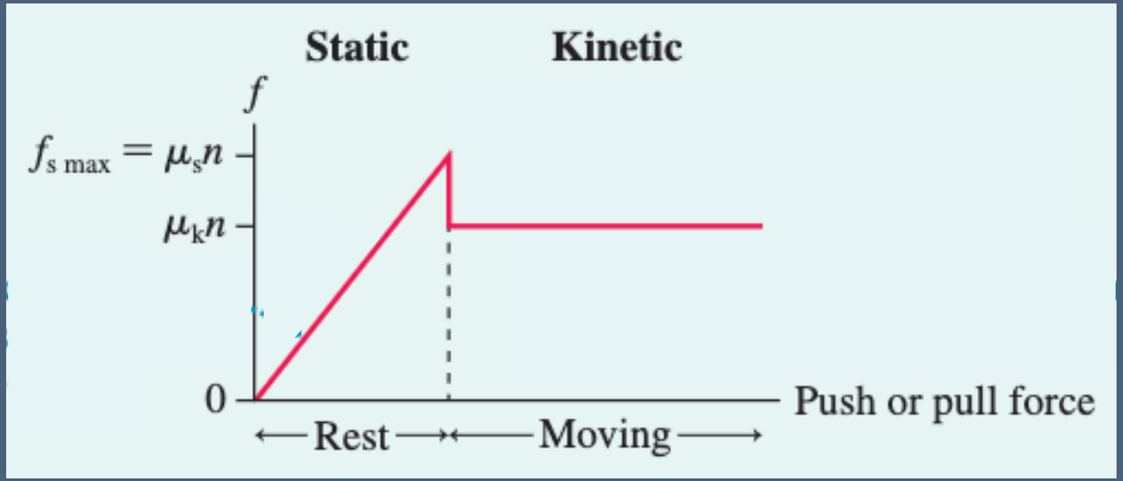
- Friction of an object moving over a surface
- Parallel with surface in direction opposite to the motion
- $f_k < f_{s\,\max}$
- $f_k = \mu_k n$ (constant)



Fiction model

- Function equations not “laws of nature”; they are a model of friction
- Friction force is parallel to the surface
- **Static friction**: Acts as needed to prevent motion. Can have any magnitude up to $f_{s\,\max} = \mu_s n$
- **Kinetic friction**: opposes motion with $f_k = \mu_k n$





Example

- Carol pushes a $25\,\mathrm{kg}$ wood box across a wood floor at a steady speed of $2.0\,\mathrm{m/s}$. How much force does Carol exert on the box? If she stops pushing how far will the box slide before coming to a rest?

- $\mu_k = 0.20$ (from table 6.1 in book)

- Draw a picture, establish a coordinate system, define symbols, identify what the problem wants

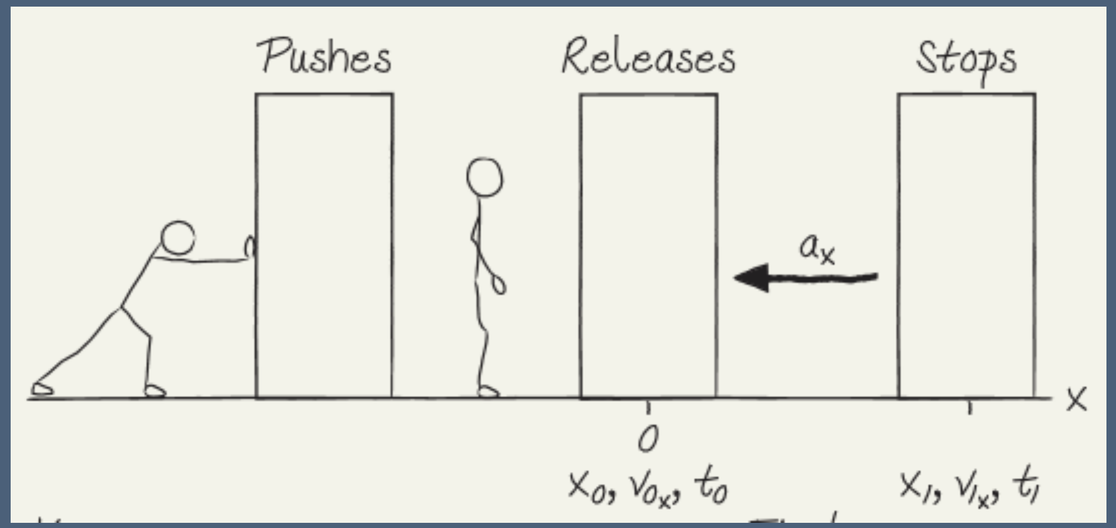
- Identify all forces acting on the object

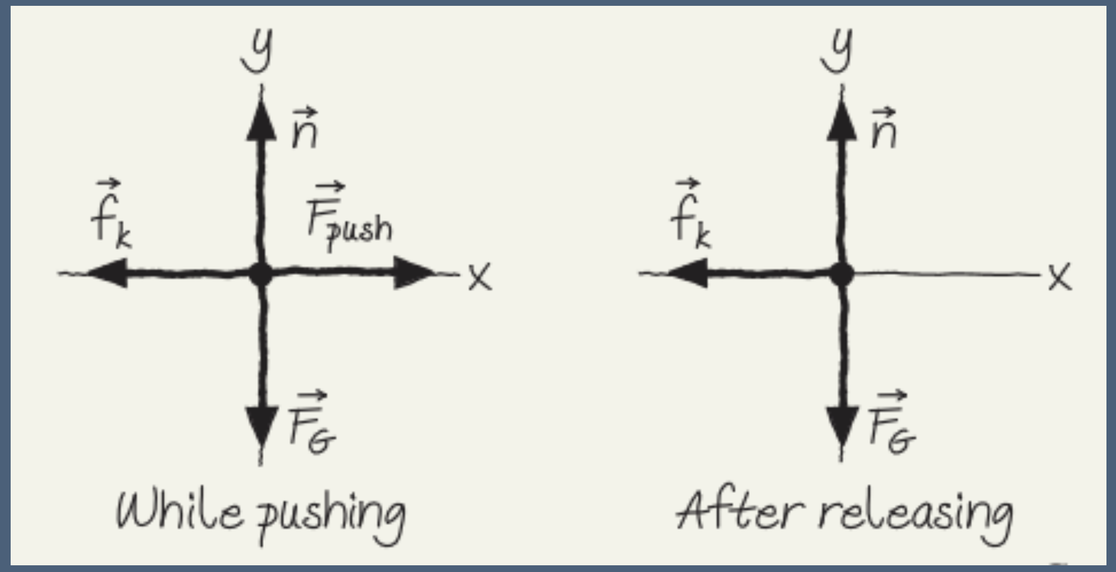
- Make a free-body diagram

- Either use $\vec{F}_{\mathrm{net}}$ to find $\vec{a}$, then use kinematics or use kinematics to find $\vec{a}$, then solve for unknown forces.

Example

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- $F_{\text{push}} = 49\,\mathrm{N}$
- $x_{\text{slide}} = 1.0\,\mathrm{m}$





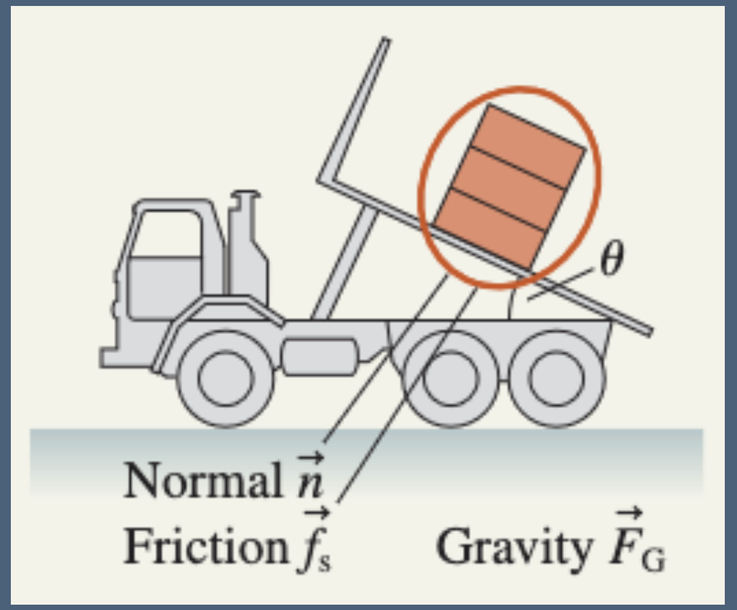
Example

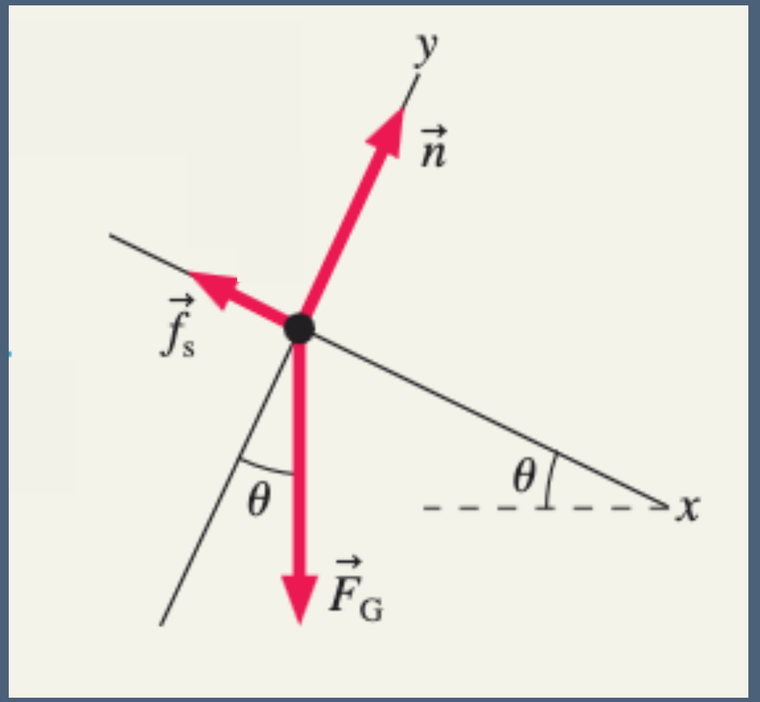
- A $50\,\mathrm{kg}$ steel file cabinet is in the back of a dump truck. The truck's bed, also made of steel, is slowly tilted. What is the size of the static friction force on the cabinet when the bed is tilted $20°$? At what angle will the file cabinet begin to slide?
- $\mu_s = 0.80$ (from table 6.1 in book)

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Example

- Compressed air is used to fire a $50\,\mathrm{g}$ ball vertically upward from a $1.0\,\mathrm{m}$ tall tube. The air exerts an upward force of $2.0\,\mathrm{N}$ on the ball as long as it is in the tube. How high does the ball go above the top of the tube? Neglect air resistance.

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- $a_y = 30.2\ \mathrm{m/s^2}$
- $v_1 = 7.77\ \mathrm{m/s}$
- $\Delta y = 3.1\ \mathrm{m}$